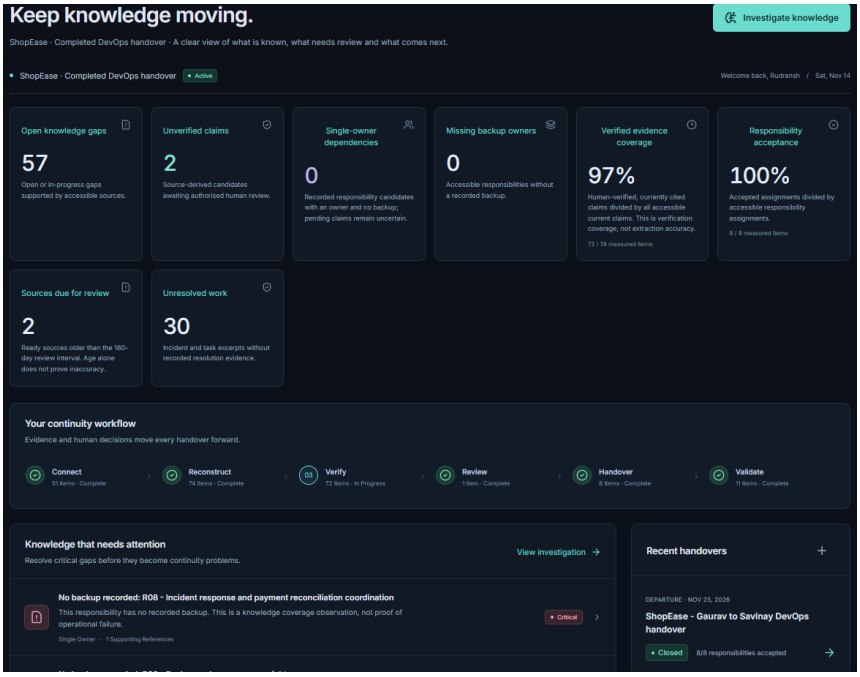
FINAL PRODUCT REPORT · ROUND 3 · TEAM JAEGERS

Knowledge that stays usable.

INHERIA

People leave. Knowledge lives on.

Technical knowledge continuity, from authorised sources to accepted responsibility and a reviewed continuation plan.



Gaurav to Savinay

A reviewed DevOps handover

OPEN THE PRODUCT

inheria.onrender.com

Live ShopEase judge workspace. Authored DevOps case study; captured 14 November 2026.

01 · Connect
Documents / GitHub

→

02 · Reconstruct
Candidates / gaps

→

03 · Verify & review
Exact sources / approval

↓

04 · Handover
Current approved version

→

05 · Accept & validate
Successor + supervisor

→

06 · Continue
Ownership + next steps

Permission and source-version checks apply at every step.

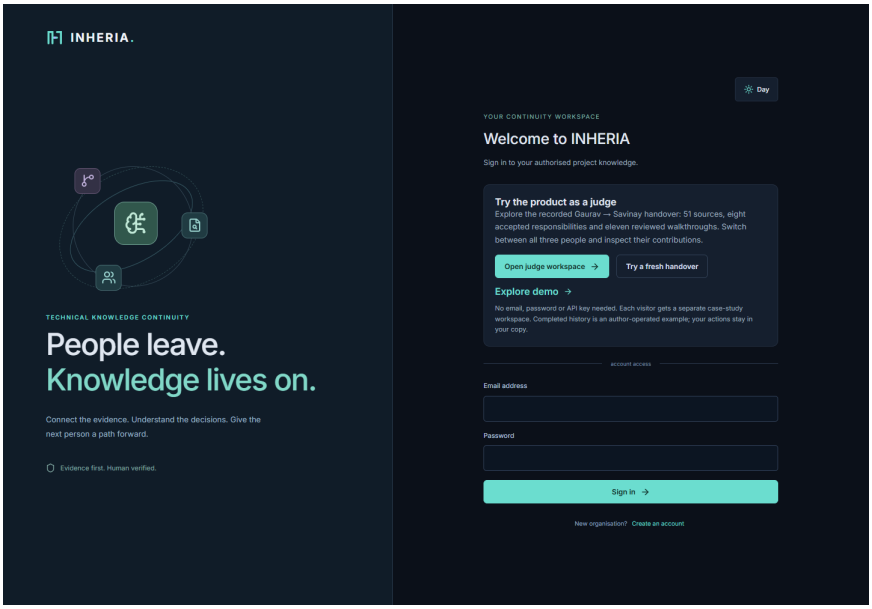
Open the working product
inheria.onrender.com
Public source code · Gauravwakle96/INHERIA
Choose **Open judge workspace** or **Try a fresh handover**. No core-workflow API key or shared password required.

Gaurav Wakle · Pranad Bhale · Savinay Pathak · Rudransh Jaiswal
MIT, Chhatrapati Sambhajinagar. Roster transcribed from the Round 1 proposal.

JUDGE QUICK START · LINKS, ROLES AND EXPECTED RESULTS

Try the product without a walkthrough.

Open the live link in a private window. The sign-in screen gives judges a complete example and a separate fresh workflow.



Entry	What the judge receives
Open judge workspace	Completed ShopEase example: 51 sources, 81 claim records, 8 accepted responsibilities and 11 reviewed walkthroughs. Inspect evidence and remaining work.
Try a fresh handover	13 authorised case-study sources and an unapproved transition. Reconstruct, review and perform the transfer yourself.
Explore demo	Scenario and role guide at /demo. Real teammates use separate accounts in one project.

Follow this short inspection

1. Open a knowledge record and inspect its source quotation, version and locator.
2. Inspect Handover, acceptance and separate supervisor review.
3. Switch to Savinay in the case-study topbar; open My continuation plan.
4. Use fresh mode to upload permitted text, run Investigation and review candidates.
5. Try a missing prerequisite: Close transition remains blocked and points to the next action.

Access and hosting expectations

Visits are separate, expiring workspaces (four-hour default). Demo role switching is restricted to that visitor. Render Free can need about a minute to wake; quota/capacity failures show a retryable message. The README also provides a verified local installation path.

Evidence: README.md · final-product-readiness.md · judge-upgrade-target-verification.json

PROBLEM, USERS AND THE CHOSEN CONTINUITY OUTCOME

A DevOps engineer leaves ShopEase.

A shopping application still needs safe releases, monitoring and recovery when its experienced DevOps engineer leaves. Files alone do not tell the successor what is current, who owns it or what remains unresolved.

Person	Role and contribution to the handover
Gaurav Departing engineer	Explains release routines, rollback restrictions, troubleshooting and implicit duties. Reviews candidate ownership and answers gap-driven questions.
Savinay Incoming engineer	Inspects approved evidence, requests missing knowledge access, accepts or rejects responsibility and submits cited validation work.
Rudransh Coordinator / reviewer	Connects sources, manages membership, arranges authorised technical review, approves the current handover and verifies closure gates.

What actually transfers?

Knowledge bundle	Relevant material and unresolved risks
Release and rollback	Pipeline configuration, runbooks, health checks, approval boundaries, schema-change constraints and deployment failures.
Infrastructure and operations	Infrastructure configuration, monitoring alerts, incident history, escalation contacts and recurring maintenance.
Backup and recovery	Backup responsibility, restore-test references, recovery prerequisites and missing or outdated recovery evidence.
Project context	Architecture, decision rationale, service dependencies, open tasks and responsibility/backup gaps.

Credible outcome

The successor receives current, reviewed knowledge and explicitly accepts ownership. Missing evidence becomes a visible task or blocker. Completion preserves open operational issues with accountable follow-up; it does not claim every incident or cloud recovery was solved.

ShopEase and its role identities are authored demonstration material, not Wipro/client operational records. Scenario actions are case-study history, not independent assessments or actual participant contributions.

Evidence: README.md · feature-status.md

CONNECT / RECONSTRUCT / VERIFY / REVIEW / HANDOVER / CONTINUE

The central workflow is complete.

01 · Connect
Documents / GitHub

→

02 · Reconstruct
Candidates / gaps

→

03 · Verify & review
Exact sources / approval

↓

04 · Handover
Current approved version

→

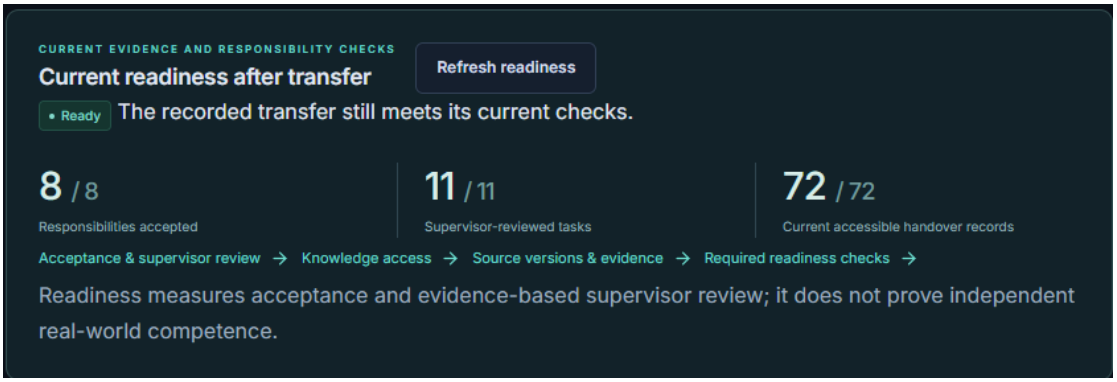
05 · Accept & validate
Successor + supervisor

→

06 · Continue
Ownership + next steps

Permission and source-version checks apply at every step.

Step	Implemented behaviour	Visible continuation result
Connect	GitHub REST/PAT; PDF, DOCX, MD, TXT, JSON ingestion.	Source status, locator, timestamp, ACL and version.
Reconstruct	Bounded source investigation and explicit English extraction.	Candidates, decisions, duties, dependencies, gaps.
Verify	Exact evidence quotes and current source checks.	Unsupported, stale and conflicting knowledge remains visible.
Review	Authorised correction, approval or rejection; employee explanations independently reviewed.	Review state and correction/audit history.
Handover	Permission-filtered report, current-version approval and proposed responsibility owner.	Role-appropriate knowledge and open work.
Accept & validate	Incoming acceptance/rejection; cited scenario submissions; separate supervisor assessment.	Recovery, reapproval and explicit prerequisite gates.
Continue	Close only when current gates pass; record accepted ownership.	Successor duties, procedures, source links and next actions.



Current readiness in the built local QA case study: eight accepted duties, eleven supervisor-reviewed tasks and 72 current accessible handover records. Changed evidence removes current readiness; these are not production-competence scores.

Evidence: API_CONTRACT.md · three-improvements-acceptance.md

PROMISE / DELIVERY / ACCEPTANCE EVIDENCE

Exactly the three submitted improvements.

The team-confirmed Round 2 commitments guide this report. Conversational chat is additional functionality and does not replace any of these three.

Committed improvement	Delivered user benefit	Recorded proof
1. Secure Team Onboarding & Access Readiness	Single-use expiring invitation, self-chosen password, project/document access request and independent decision, revocation and audit.	7 dedicated journeys; live access checks; PostgreSQL invitation/permission races.
2. Investigation & Source-Change Intelligence	Explicit prose owner/rationale/dependency extraction; cited downstream impact; revalidation after change/removal; source-version checkpoints and resume.	60 labelled docs (34 prose) initial internal holdout; 10 distinct changes/revocations; 5 interruption positions. Separate later first-run evaluation.
3. Practical Transfer Validation & Recovery	Deployment/rollback/recovery rubrics; cited staging records; correction/reopening; critical prerequisites and traceable reapproval.	12 positive/negative role flows; complete UI transfer; current and revoked observation evidence checks.

239
Backend tests passed

17
Real PostgreSQL cases

34/34
Exact-three journeys

Delivered scope and measured limits
The dedicated acceptance runner passed 34/34 journeys. Internal evaluation labels are implementation-informed. External staging authenticity and operational competence need authorised records and qualified supervisors. INHERIA executes no production commands.

Changes from the broad Round 1 vision

The eleven-engine proposal was narrowed to the technical source-to-continuation journey. Existing components were extended. Industry profiles, workflow builder, cross-company learning, predictive scores and broad autonomous remediation were deferred. PostgreSQL, lexical retrieval and bounded workers keep the deployment understandable and affordable.

Evidence: three-improvements-acceptance.md · three-improvements-tests.json · round3-form-answers-draft.md

SECURE TEAM ONBOARDING & ACCESS READINESS

1 · A usable handover starts with access.

Responsibility assignment does not grant knowledge access. The successor must be able to read the current documents supporting the handover.

Mechanism	Implemented safeguard
Invitation	Random hashed, expiring, single-use token. New recipients set a password; existing recipients authenticate without role/password overwrite.
Access request	Incoming user requests missing knowledge access. A separate authorised source reviewer approves or rejects with a note. Unreadable source titles stay hidden.
Document permissions	Current reader list applies to source text, derived claims, graph, handover, saved history and readiness. Admin does not bypass a source ACL.
Revocation	Project member removal revokes pending requests/invitations and explicit reader grants. Last-reader removal never opens access to everyone.
Audit and concurrency	Invitation/member/access decisions are traceable. PostgreSQL project-row locks and fresh ORM checks prevent stale acceptance/revocation races.

A judge can exercise this improvement

Try it
Knowledge owners / Invite member / privately open the one-time link / choose a password. Restrict a required source, inspect missing access as the successor, request access and review it as an authorised reviewer. Revoke membership and check denial.

Acceptance cases	Outcome and scope
7 dedicated journeys	Single-use, expiry, independent review, missing-access recovery, membership denial, document-reader updates and last-reader safety.
Live + PostgreSQL	Browser invitation/source-reader checks and bounded overlapping database writes. No enterprise identity certification.

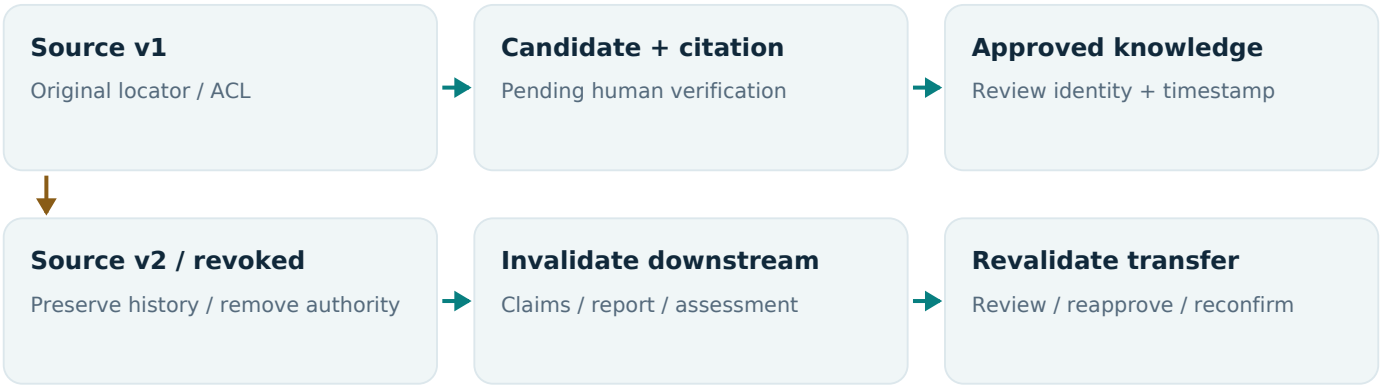
Limit: links are manually shared. Email delivery, verified email, password reset, SSO/MFA and external AWS/IAM grants remain future work. Project removal does not mean organisation-wide offboarding.

Evidence: three-improvements-acceptance.md · test_onboarding_completion.py · test_source_access_completion.py

INVESTIGATION & SOURCE-CHANGE INTELLIGENCE

2 · Source changes cannot leave false readiness.

Knowledge keeps its provenance through extraction, review and transfer. A changed, deleted or revoked source removes current authority from dependent work.



Delivered capability	Practical effect
Explicit prose extraction	Owner, rationale and dependency spans remain exact-cited candidates. Historical/negated/training-example ownership is excluded; absent rationale stays absent.
Cited dependency impact	Graph relationships and cycle-safe paths link affected systems, duties, knowledge and transfer work to accessible evidence.
Change/revoke/delete	Preserve version history, invalidate dependent claims/assessments, withhold inaccessible evidence and purge source-derived text on deletion.
Checkpointed recovery	Candidate writes and source-version checkpoint commit atomically. Authorised resume skips unchanged checkpoints and avoids duplicates.

Acceptance delivered

10 distinct changes/revocations: replacement, revoke, deletion, successor ACL loss, transitive interview invalidation, GitHub 401/403/404, definitive tree-file removal and README removal. Five interruption positions (2, 3, 4, 7, 11) resume without duplicate candidates. A separate actual process-restart test also passed.

Initial submitted evaluation: 60 separately authored documents, including 34 prose cases, met the 90% precision and 85% field targets on internal labels. Page 12 gives the later first-run evaluation, denominators and preserved failures.

Limit: explicit English patterns and bounded in-process work. This is not unrestricted semantic understanding or a distributed durable job system.

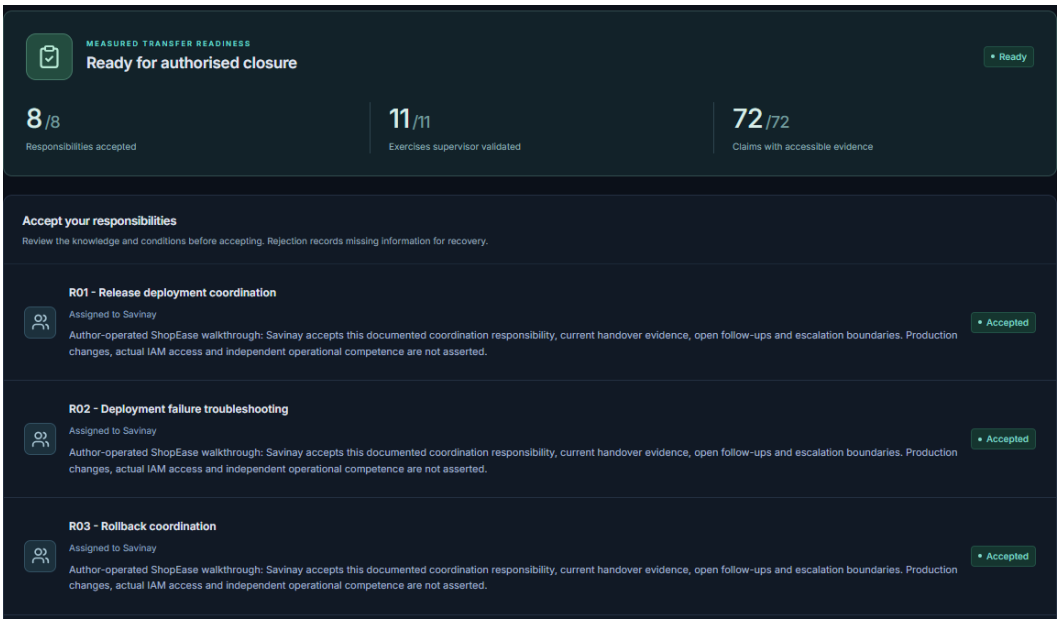
Evidence: three-improvements-tests.json · judge-input-hardening.md · extraction-holdout-report.json

PRACTICAL TRANSFER VALIDATION & RECOVERY

3 · Prove transfer, recover incomplete work.

Savinay submits a cited explanation or supplied staging observation. A separate authorised supervisor reviews the relevant criteria before closure.

Scenario	What the successor must demonstrate
Deployment review	Identify the current pipeline, prerequisites, approval boundary, post-release checks and escalation when evidence is missing.
Rollback readiness	Locate approved steps, identify schema/data compatibility restrictions and explain when specialist review is required.
Recovery evidence	Locate backup/restore evidence and accountable owner. A runbook alone is not accepted as a staging restore observation.



Cropped live case-study supervisor view. Documentation walkthroughs and externally supplied staging observations are distinguished; source versions and assessment history remain inspectable.

Recovery is part of the workflow

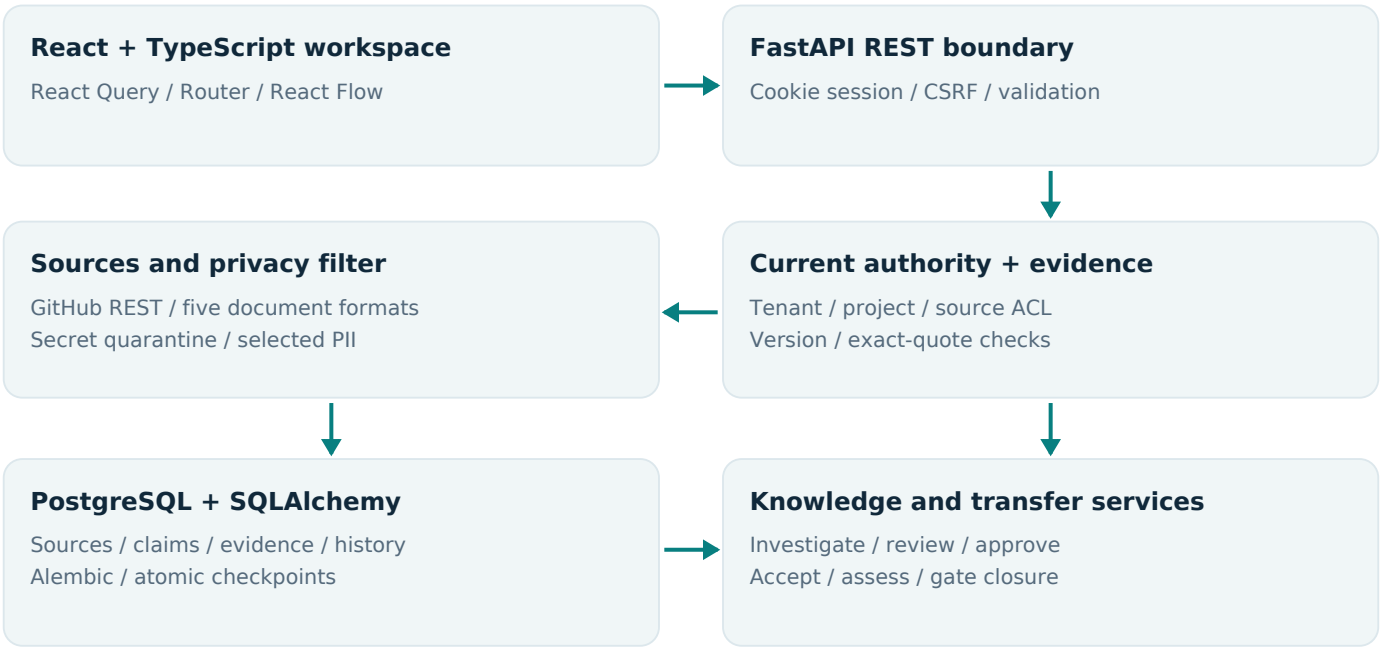
Reject incomplete ownership / request corrections / resubmit / reassess. Reopened active handovers require a new report, reapproval, renewed acceptance and assessment. Missing critical prerequisites or changed/revoked evidence block closure. Closed ownership remains history; later changes use a follow-up transition.

12 role flows passed: correct/wrong acceptance, missing citation, wrong submitter, self-assessment denial, rejection, incomplete rubric, correction, critical prerequisite, reopening, staging observation and revocation. External observation authenticity still needs qualified review.

Evidence: test_transfer_acceptance.py · three-improvements-acceptance.md

TECHNOLOGY, DATA FLOW AND IMPORTANT DECISIONS

Small architecture, explicit trust boundaries.



Layer	Stack and consequential choice
Frontend	React 19, TypeScript, Vite 7, React Query, Router and React Flow. One responsive workspace; same-origin REST in production.
Backend	Python / FastAPI / Pydantic / httpx. Modular monolith with domain rules separate from UI and provider adapters.
Persistence	SQLAlchemy 2, PostgreSQL/psycopg in production; SQLite local default; immutable Alembic migrations. UUIDs, foreign keys, indexes and uniqueness constraints.
Processing	PyMuPDF, python-docx and bounded GitHub REST. Database-backed job state, atomic source checkpoints and in-process workers.
Hosting	Docker builds UI and API; one Render Free HTTPS origin plus Neon Free PostgreSQL. No separate graph/vector database or paid worker.
Optional AI	Provider adapters validate structured outputs and accessible citations. Rule-based core and retrieval remain usable without a model key.

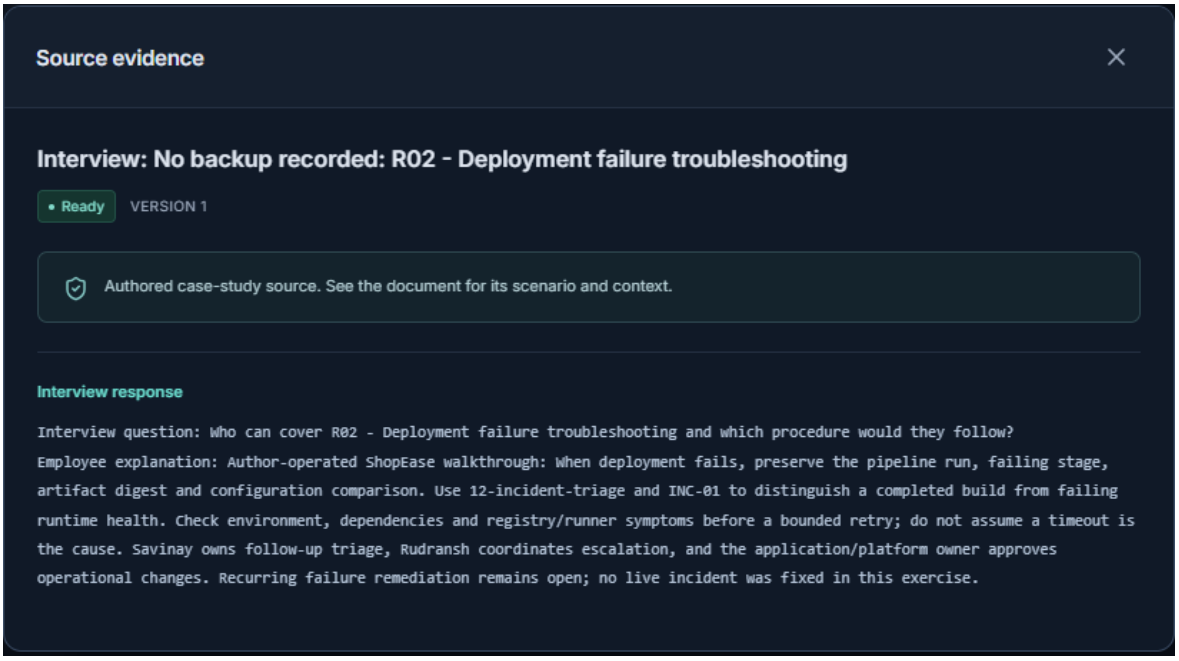
Hard engineering boundary

Current permissions and evidence versions are rechecked across extraction, reports, saved conversation, acceptance and practical assessment. A cached or historic source cannot bypass revoked authority. GitHub duplicate enqueue and invitation races use project locks plus refreshed state.

Evidence: architecture.md · API_CONTRACT.md · pyproject.toml · package.json

DATA MODEL, CITATIONS, REVIEW AND EXPLAINABLE METRICS

Every important statement has a lineage.



Cropped original-source modal from the live case study. Employee explanation remains identified as testimony and requires independent review.

Persisted domain	Records and invariants
Identity and scope	Organisations, users, membership, sessions, invitations, demo visits and access requests. Tenant/project/source checks precede content access.
Knowledge evidence	Connections, sources, versioned chunks, claims and evidence. Citation retains source ID/name/type/time/version, chunk, locator and exact quote.
Investigation and change	Investigations, gaps, simulations and jobs. Current authority, source-version checkpoints and derived-evidence lineage.
Transfer and audit	Transitions, transfers, exercises and audit events. Versioned report approval, actor-specific acceptance, separate assessment and historical ownership.
Optional conversation	Encrypted organisation configuration, owner-only conversations, turns and expiring action previews. No private key returned.

What the dashboard numbers mean

Acceptance = accepted duties / required duties. **Validation** = supervisor-reviewed passing tasks / required tasks. **Evidence coverage** = current readable handover records / required records. Gaps, stale sources and unresolved issues are counts with source-backed definitions. Missing denominators yield unavailable values; these are not AI-generated competence scores.

Evidence: models.py · API_CONTRACT.md · architecture.md

AUTOMATED, DATABASE, CONNECTOR AND END-TO-END EVIDENCE

Reliability was exercised at the boundaries.

239
Backend tests passed

17
Real PostgreSQL cases

34/34
Exact-three journeys

Recorded check	Result	What it establishes
Latest backend suite	239 passed; 0 failed/skipped	17 real PostgreSQL cases in owned loopback QA schemas. Latest run did not collect coverage; previous measured baseline 89%.
Exact-three acceptance	34/34 passed	7 onboarding + 10 distinct source changes + 5 interruption positions + 12 role flows. Isolated fixtures/mocked failures.
Final uncertainty guard	22 targeted passes	Boundary rerun after the final claim/assignment uncertainty change. No new full-suite coverage claim.
Focused browser checks	9/9 passed	Readiness errors, cooldown, access refresh, source warnings, layout and active job completion/failure.
Complete local UI journey	Passed ~203 sec	Fresh sources / review / report / ownership acceptance / separate assessment / closure. Author-operated automated QA.
Real live GitHub sync	40 initial imports; 0 new on resync	Own public repository, authenticated bounded sync; temporary token revoked. Not arbitrary private-repository proof.
Clean installation	Passed	Fresh Git archive, frozen Python install, npm ci/build, migration, API start, 13-source fresh judge entry and deep link.
CI and dependency checks	Passed	Backend lint/format/tests, frontend lint/typecheck/build, Python/Node audits and scoped secret scan.

Failures found and corrected

Reproduced stale invitation revocation and duplicate-sync races; fixed lock/reload sequencing. Fixed mobile source-table overflow. Request-limit errors show cooldown and preserve saved work. Missing/failed readiness disables closure. Source changes and access loss require recovery rather than silently retaining success.

Latest successful CI run · Live application commit

Evidence: judge-upgrade-verification.json · judge-upgrade-target-verification.json · github-live-sync-verification.json

EXTRACTION QUALITY · INTERNAL FIRST RUN AND REGRESSION LIMITS

Evaluation reports the misses as well.

The latest separate first-run corpus has 60 new prose documents, different identities and systems, frozen after general fixes and before execution. It was not tuned after this run. Labels are authored internally with implementation knowledge.



Amber marker = submitted target. Scale: 0 to 100%.

Metric	Latest separate first run	Submitted target
Candidate precision	50/50 = 100%	At least 90%
Candidate recall	50/52 = 96.2%	Reported; no promised threshold
Owner accuracy	19/20 = 95%	At least 85%
Rationale accuracy	11/12 = 91.7%	At least 85%
Dependency fields	20/20 = 100%	At least 85%
Exact quotations	50/50 = 100%	Attribution integrity; not semantic truth

Two conservative misses remain

An unnamed engineer duty with a separate denial was not emitted. A definite decision with a rationale containing “may” was also missed. Missing knowledge remains unresolved; the product does not invent an owner or rationale.

Original failures remain published

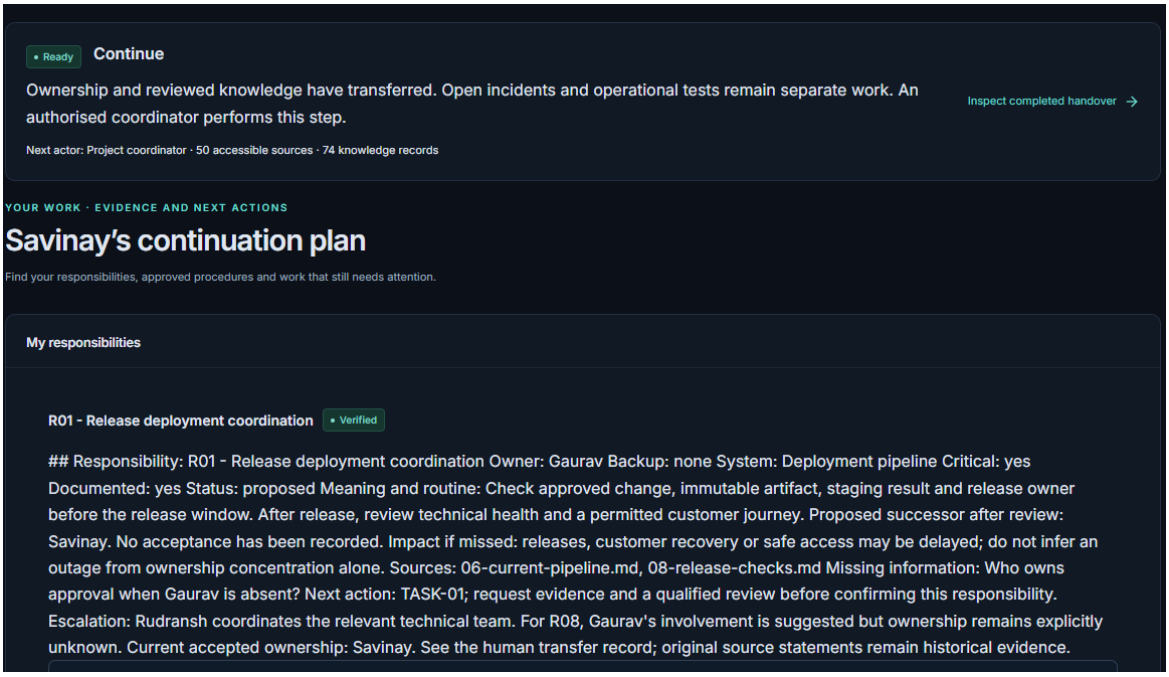
A prior frozen 72-prose challenge first measured precision 95.2%, recall 71.4%, owner 65%, rationale 50% and dependencies 20%. General fixes now pass that known set at 100%; those results are **tuned regression**, not fresh accuracy. The initial submitted 60-document / 34-prose internal holdout is retained separately.

Scoring uses exact candidate/field identities and normalises the leading article “the” in dependency endpoints. These bounded sets do not establish independent enterprise or multilingual accuracy. Independent labels, representative documents and reviewer agreement are roadmap gates.

Evidence: postfix-internal-first-report.json · judge-input-first-report.json · judge-input-regression-report.json

USABILITY, ACCESSIBILITY AND REAL-WORLD FIT

The next person can find the next action.



Cropped live Savinay continuation view: current duties, reviewed procedures and source inspection in the same workspace.

Design decision	How it helps a new user
Visible next actor/action	The current project state computes where the coordinator or successor should go next; empty and missing states explain what is needed.
Closure before action	Readiness shows acceptance, independent review, current evidence and blocker links before Close. Backend gates are authoritative.
Recoverable failures	Safe retry/cooldown, preserved saved data and manual readiness/access refresh make failure and teammate updates understandable.
Readable evidence	Sources, versions and locators remain inspectable. Uncertainty is explicit; role-specific reports omit restricted material.
Responsive interface	390 / 768 / 1440 pixel checks; day/night themes, keyboard controls and reduced-motion support. Bounded axe checks passed.

What user validation has actually occurred

Automated browser journeys and author-operated live checks are recorded. No unfamiliar-user observations were supplied, so this report makes no five-person usability-study claim. The repository includes a consent-based task pack to record completion, time, help, confusion and retest outcomes.

Remaining accessibility work: complete screen-reader and assistive-technology testing with users; bounded automated checks are not a full WCAG certification.

Evidence: usability-task-pack.md · judge-experience-verification.json · capture-record.json

PRIVACY, SECURITY AND RESPONSIBLE USE

Permission-aware by design.

Trust boundary	Implemented protection	Practical limit
Identity	Argon2 passwords; opaque hashed sessions; Secure/HttpOnly/SameSite cookies.	No verified email, reset email, SSO or MFA.
Server writes	Pydantic validation, cookie/CSRF/Origin checks; parameterised ORM queries.	No independent penetration-test certification.
Content disclosure	Tenant/project/source ACL plus evidence-version checks on derived views/history.	Upstream source ACL mirroring is absent; resync/revoke required.
Sensitive sources	Pattern-based secret quarantine and selected contact-PII redaction before indexing.	False positives/negatives possible; no comprehensive DLP.
Credentials	Fernet-encrypted provider/PAT keys; environment secrets; tokens excluded from UI/audit.	Operator must protect encryption key and assess provider agreements.
Review and action	Candidates pending; correction/rejection/approval; separate successor/supervisor acts.	Human approval does not establish external operational truth.
Lifecycle	Version history, derived invalidation, source deletion/purge and filtered audit.	Automated retention/legal hold/data-residency policies not implemented.
Demonstration	Separate expiring visitor tenant; private credentials and conversations excluded.	Do not upload confidential production data into a public demo.

Safe operation is explicit

INHERIA handles knowledge and responsibility transfer. It records authorised staging observations and supervisor review; it does not run production deployment, rollback or restore commands. Hypothetical risk analysis identifies assumptions. Unsupported generated content is never treated as verified organisational truth.

External services and licence considerations

GitHub, Render and Neon follow their own access/usage terms. React, FastAPI and the listed ecosystem tools use their respective open-source licences; Inter and bundled Bitstream Vera have redistributable font licences. **PyMuPDF uses AGPL/commercial licensing**: review the intended distribution/deployment obligations. Model processing needs organisation opt-in and provider-retention review.

Evidence: architecture.md · deployment.md · pyproject.toml

REPRODUCIBLE SETUP · ENVIRONMENT · API · OPTIONAL AI

Clone, install, run and verify.

Verified fresh installation: Python 3.13, uv, Node 24 and npm. Metadata supports Python 3.12+. Locked dependencies define the actual installed versions.

PowerShell · repository and API

```
git clone https://github.com/Gauravwakle96/INHERIA.git
cd INHERIA/backend
uv sync --frozen
Copy-Item .env.example .env
uv run alembic upgrade head
uv run uvicorn app.main:app --host 127.0.0.1 --port 8010
```

For an existing checkout, preserve the current .env. In a second terminal, from the root:

PowerShell · frontend

```
cd frontend
npm ci
npm run dev -- --host 127.0.0.1 --port 5173
```

Open http://127.0.0.1:5173. API docs: http://127.0.0.1:8010/docs.

Environment variable	Purpose
INHERIA_DATABASE_URL	SQLite local default; PostgreSQL/psycopg with TLS in production.
INHERIA_ENVIRONMENT	Production requires PostgreSQL and HTTPS.
INHERIA_TOKEN_ENCRYPTION_KEY	Stable server-side Fernet key for private connector and AI credentials.
INHERIA_ALLOWED_ORIGINS	Exact permitted origins. Never wildcard private authenticated writes.
INHERIA_DEMO_MODE	Shared development fixture access; false in production.
INHERIA_JUDGE_DEMO_ENABLED	Separate visitor demo; four-hour expiry and forty-visit default.
INHERIA_LLM_ENABLED	Optional structured extraction opt-in; disabled by default.

Optional AI: bring your organisation’s own key

Administrator / Digital successor / AI settings. Bynara uses **https://router.bynara.id/v1**, **agnes-2.5-flash** and Chat Completions; existing AgentRouter options remain. Enter the key privately in settings, enable and test. No key is supplied to judges. Provider usage can incur charges; core ingestion/review/transfer needs no AI key.

Verification: backend `uv run pytest -q`; exact-three runner `uv run python evaluation/run_submission_acceptance.py`. PostgreSQL cases require the documented owned loopback QA URL. Frontend: `npm run lint`, `npm run build` and configured Playwright E2E. See README for test-server configuration.

Evidence: README.md · .env.example · API_CONTRACT.md · deployment.md

COMPLETE FEATURE INVENTORY · WHAT REMAINS OUTSIDE THE PRODUCT

Finished features and honest boundaries.

Working feature group	Delivered functions
Accounts and access	Registration/login, project membership, expiring invites, password choice, access requests, source readers, revocation and audit.
Sources	GitHub REST/PAT, five document formats, provenance/locators, incremental sync, retries/warnings, deduplication, replacement, revoke and delete.
Intelligence	Project-state candidates, explicit English extraction, decisions/rationale/alternatives, dependency graph, ownership gaps, targeted text capture and risk scenarios.
Human trust	Exact evidence inspection, uncertainty, independent correction/approval/rejection, history, source-change impact and resumable investigation.
Transfer	Versioned report, approval, successor/backup assignment, acceptance/rejection, deployment/rollback/recovery criteria, cited observation, supervisor review and recovery gates.
Continuity	Current ownership, completed/fresh demo, continuation plan, contributor/history records, next-step guidance and explainable metrics.
Interface and delivery	Day/night, responsive/keyboard/reduced motion, filtered privacy/audit, migrations, Docker, free deployment, CI and reproducible tests.
Additional optional AI	Private conversations, encrypted organisation key, Low/High budgets, supported model selection, citations, labelled retrieval fallback and confirmed action previews.

Known limitations that a judge may encounter

Scanned PDFs need external OCR. Prose patterns cover explicit English; shared ownership/pronouns and broad semantics remain incomplete. GitHub imports bounded recent history/selected files, not a full repository archive. No OAuth/webhooks, voice capture, general workflow builder or multi-successor split.

Render cold starts, Neon storage/transfer quotas, shared-IP limits and the forty-visit cap constrain capacity. Latest pre-check transfer snapshot: about 4.49 GB used of 5 GB; remaining quota is not guaranteed. Jobs run in-process; no distributed workers or scheduled retention. No SSO/MFA/reset email or enterprise governance.

Optional chatbot reliability

Live Bynara replies are recorded, but a prior 50-question provider run returned only 9 valid replies and 41 labelled fallbacks. That is provider-limited functionality, outside the three submitted improvements. Exact citation structure does not prove semantic correctness.

Evidence: feature-status.md · verification.md · four-workstream-acceptance.md

PRIORITIES · DEPENDENCIES · EFFORT ESTIMATES · RELEASE GATES

Product roadmap and delivery strategy

Four milestones deepen the continuity workflow. Effort ranges are small-team planning estimates, not release dates. These are future capabilities; the three submitted improvements remain the delivered commitments.

Milestone / estimate	User benefit and delivery	Dependency / release gate
M1 · Independent quality P0 · 2–3 development weeks	Independent permissioned labels, category quality and better owner/rationale/dependency extraction. Reviewers know where correction is needed.	Consent and label agreement precede changes. Untouched evaluation meets ≥90% precision and ≥85% field targets with misses disclosed.
M2 · Identity and onboarding P0 · 3–4 development weeks	Verified email, secure reset, approved identity provider and accessible role-guided onboarding. Successors join/recover access safely.	Freeze ACL contracts; email verification before reset/SSO. Permission/replay/concurrency tests and observed unfamiliar-user journeys.
M3 · Jobs and recovery P1 · 3–4 development weeks	Durable jobs, shared limits, scheduled review/retention and rehearsed application restore. Operators recover expected failures.	Measure workload/budget before queue choice. Restart/duplicate/revoke gates, approved RPO/RTO and published capacity results.
M4 · Permissioned pilot P2 · 4–6 development weeks per pilot	One needed connector, specialist review, staging artefacts and domain schemas for a real consenting team.	After M1–M3: real auth/sync/revocation/E2E and pilot feedback. No production remediation or fabricated operations.

Delivery ownership and risk control

Product coordination owns priority; data/AI engineering owns quality; backend/security owns access; platform engineering owns recovery; QA/UX owns release evidence. These are planned workstreams. Consent, identity policy, reviewer availability and hosting costs are dependencies, not assumed resources.

Milestone definition of done

Ship user-visible scope, API/schema/environment contracts, permission/failure tests, appropriate user observations, migration/recovery instructions, licence/cost disclosures and clean setup. A failed gate delays expansion. Future targets are never presented as measured results.

Evidence: roadmap.md · operating-guide.md

DETAILED ROADMAP · DATA QUALITY · ATTRIBUTION · ACCEPTANCE

M1 · Independent reconstruction quality

Reviewers need reliability evidence for their sources. Current explicit English patterns and internal labels remain bounded; M1 establishes independent evidence before wider semantic claims.

Work package	Planned implementation and artefact
Permissioned corpus	Consent/redaction/retention register; proposed ≥100 documents from ≥2 categories, including ≥40 prose-heavy and ≥20 ambiguous/negative cases.
Independent labels	Two reviewers label type, owner, current/historical state, rationale, dependency direction and exact span. Publish disagreement and adjudication.
Frozen evaluation	Split by project/document lineage before changes. Separate training, tuned regression and untouched test data; publish manifest/hash.
Extraction depth	Scoped pronouns, shared/unnamed ownership, causal reasons and multiword dependency fields. Missing evidence remains unresolved.
Provider comparison	Same cases/ACLs, timeout and cost budget for rules/optional providers. Keep useful non-provider review available.

Measurable release gates

Measure	Proposed gate and denominator
Precision	≥90%: correct candidates / extracted candidates on untouched labels.
Field accuracy	≥85% each owner/rationale/dependency: correct / applicable labels; report category failures.
Recall / abstention	Publish recovered / relevant candidates by category and conservative misses; do not conceal unanswered cases.
Attribution / permission	Valid current spans; zero restricted/revoked disclosures across API, graph, reports and history acceptance cases.

Dependencies and risk response

Independent reviewers and source consent can block the corpus. Keep internal labels disclosed and the bounded release usable; never rename them independent. Ambiguous ownership may remain unassigned. Optional provider performance does not block core review.

Release packet: permission register, label guide, agreement/adjudication, frozen manifest, first-run category results, preserved misses and regression suite. Estimate: 2–3 development weeks after source/reviewer availability.

Evidence: roadmap.md · postfix-internal-first-report.json

DETAILED ROADMAP · SECURITY · USER RESEARCH · ADOPTION

M2 · Identity and accessible onboarding

Successors should join safely, recover credentials and understand missing access with minimal help. Extend current invitations and readiness while preserving tenant/source authority.

Sequence	Implementation and dependency
1 · Verify / deliver	Verified-email registration and invite delivery; hashed expiring one-use links, enumeration protection and resend/accept limits.
2 · Recovery	Password reset, invalidation and notification. Test expiry/replay, simultaneous use and member revocation against PostgreSQL.
3 · Identity	One approved identity provider/MFA policy; tenant binding, retained memberships and no silent account merging/role escalation.
4 · Source authority	Justified OAuth/signed webhooks; signature/replay, expiry, repository removal and permission-loss handling.
5 · Observed use	≥5 consented unfamiliar-user sessions across three roles; keyboard, focus, low-vision and representative screen-reader journeys.

Acceptance must demonstrate usable access

Security gates
Invite/verify/login/reset/expiry/replay/concurrency/revoke flows pass. Forbidden titles, text, citations and history stay hidden after access loss. Admin does not override source ACL; INHERIA membership grants no external AWS/IAM access.

Usability gates
Proposed target: ≥4 of 5 participants complete the assigned main task without facilitator help. Record task/time/help/confusion and unsuccessful outcomes. Fix/retest every blocker. Publish keyboard/screen-reader scope and limits; no certification inferred.

Dependencies: approved email provider, identity policy and participant consent. Test one configuration first and retain existing login through migration. Estimate: 3–4 development weeks after provider/policy access.

Release packet: threat model, provider guide, permission/replay/concurrency results, audit examples, actual user observations, accessibility scope and migration/recovery instructions.

Evidence: roadmap.md · usability-task-pack.md

DETAILED ROADMAP · RELIABILITY · GOVERNANCE · CAPACITY

M3 · Durable jobs and recovery

Restarts, duplicate delivery and quota pressure must preserve one logical result and current authority. Infrastructure follows measured workload and an approved budget.

Work package	Planned capability
Measure first	Document counts, sync duration, DB transfer, memory, queue depth and API latency determine worker requirements.
Durable execution	Transaction-safe enqueue, stable job IDs, atomic checkpoints, bounded retries/backoff, cancellation and dead-letter recovery.
Governance	Shared limits/concurrency, permission versus transient failure, scheduled source freshness and retention/derived purge.
App recovery	Operator-approved RPO/RTO and backup scope; owned test restore checks rows, tenant ACLs, source versions and transfer state.

Failure acceptance matrix

Failure	Required result
Restart / duplicate	Committed progress and one logical artefact set; no duplicate job/source/claim.
Revoke during retry	Recheck source authority; revoked knowledge never becomes current through replay.
Budget / cancellation	Explicit terminal state/recovery; no infinite retry or partial success label.
Purge / restore	Derived purge verified; test RPO/RTO met without tenant leakage or stale-evidence revival.

Capacity target and limits

Proposed test profile: 10 concurrent users + 3 active investigations for 30 minutes. Publish hardware, inputs, p95 latency, failure rate, transfer and memory; select support limits after measurement. This is a future profile, not achieved throughput.

Risks: queue services may exceed free budgets; retain bounded mode until justified. Restore the encryption key alongside encrypted integrations. Database downgrade remains separately reviewed. Estimate: 3–4 development weeks. Release packet: failure-injection, retention/restore drill, capacity and operator/migration reports.

Evidence: roadmap.md · operating-guide.md

DEPLOYMENT · RECOVERY · SUPPORT · DOCUMENTATION CONTRACT

Operating guide and release maintenance

An administrator needs recovery instructions as well as installation commands. These procedures describe the current bounded product; future automation remains in the roadmap.

Expected condition	Safe operator or user response
Cold start / quota	Allow service wake-up; retry when offered. Monitor quota without keep-alive or extra-project bypass.
Missing permission	Successor requests knowledge; authorised independent reviewer checks ACL. External IAM stays separate.
Blocked closure	Follow readiness links; resolve acceptance, assessment, current evidence and prerequisites; refresh.
Interrupted / duplicate	Authorised investigator resumes checkpoint. Wait on 409 existing job; avoid repeat enqueue.
429 / provider failure	Honour cooldown and safe connector warnings. Optional AI preserves history and labelled retrieval.
Changed / rejected	Reinvestigate/review, regenerate/reapprove and reassess active transfer. Closed ownership uses follow-up.

Deployment and migration checklist

1. Locked dependencies, reviewed additive migrations and approved secrets.
2. Production PostgreSQL/HTTPS, stable encryption key and disabled shared development login.
3. Verify /api/health, explicit /api/ready, SPA deep link, login, CSRF rejection, ingestion and restricted read.
4. Application rollback requires schema compatibility; database downgrade/restore is separately reviewed.
5. Maintain operator recovery/key storage and quota monitoring.

Documentation definition of done
Every release updates scope, API/schema/environment contracts, recovery steps, permission/failure tests, appropriate user observations, licence/service costs and limits. Recheck clean install and retain original evaluations. Source ACL and citation lineage are release invariants.

M4 pilot gate: after M1–M3, choose one consented source-system need. Real auth/sync/permission-loss/E2E and qualified supplied staging evidence precede expansion. Estimate: 4–6 development weeks per pilot. No production command execution implied.

Evidence: operating-guide.md · deployment.md · roadmap.md

TEAM JAEGERS · CONTRIBUTION ATTRIBUTION · DOCUMENT CONTROL

Team and submission responsibilities

Team and institution are transcribed from the existing Round 1 proposal: JAEGERS, MIT, Chhatrapati Sambhajinagar. These are participant names, separate from the ShopEase case-study roles.

Participant	Assigned workstream in Round 1	Actual contribution evidence
Gaurav Wakle Team leader	Product scope/architecture, backend/AI integration and overall delivery.	Leader confirmation required for final individual contribution attribution.
Pranad Bhale	Backend/data integration: GitHub, ingestion, provenance and connector tests.	Assigned role is documented; personal delivery is not independently inferred.
Savinay Pathak	Frontend/UX: workspace, evidence, review, handover and workflow usability.	Assigned role is documented; final actual contribution needs team sign-off.
Rudransh Jaiswal	Security/QA/documentation: permission flows, edge cases, setup and demo reliability.	Assigned role is documented; final actual contribution needs team sign-off.

Document control and verification scope

Application and test references identify the exact evidence supporting this report. Proposed roadmap milestones remain separate from completed capabilities. Assigned workstreams are documented; actual individual delivery requires team confirmation.

Snapshot	Public record
Verified application	55b11e1 · input, sync and source-health hardening
Evidence snapshot	9ed1d05 · deployment and clean-install record
Latest CI at report build	Run 43612957597 · success
Source repository	Public source and documentation

Items the team leader must finalise

Confirm individual contributions and organiser-required registration details. Publish the final PDF with judge access, test submitted links in a private window and submit once for the whole team. This report does not claim the form has been submitted or personal confirmation boxes ticked.

Participant delivery and user observations require actual supplied records. ShopEase case-study identities do not establish personal team contributions. The three submitted improvements are mapped on pages 5–8.

CLICKABLE REFERENCES · REPRODUCTION · FINAL PACKAGE CHECK

Evidence index and documentation map.

Judge question	Primary public evidence
Can I use the product?	Live product · Judge guide · README
What exactly was delivered?	Three-improvement promise map · 34-journey record
How does it work?	Architecture/trust boundaries · API contract · Models
Which tests were run?	Verification timeline · Latest 239-test/9-browser record
Does clean install work?	Live + clean-install evidence · Deployment/setup limits
Was GitHub actually exercised?	40 imports / unchanged resync / revoked token
Is extraction quality measured?	Separate 60-prose first run · Original 72-case misses · Known regression
What remains incomplete?	Feature status · Detailed roadmap · Usability task pack
Can I reproduce the evidence?	Backend locked dependencies · Frontend lockfile · CI pipeline

Where the Round 3 criteria are addressed

Criterion	Report section
Completeness · 25	Pages 2–4: judge access, users and working central journey; page 16: feature inventory.
Three improvements · 20	Pages 5–8: exact commitment, implemented behaviour, demonstration and acceptance evidence.
Reliability · 20	Pages 11–12: database/connector/UI tests, evaluation denominators, failures and limits.
Usability and fit · 20	Pages 2–3 and 13–14: next steps, permissions, evidence inspection, accessibility and responsible use.
Documentation / roadmap · 15	Pages 9–10, 15, 17–23: architecture/data, setup, detailed milestones, operations, team attribution and references.

Submission package

Working product link, public source repository, this report, completed features (page 16), architecture (page 9), exact improvements (pages 5–8), testing (pages 11–12), detailed roadmap (pages 17–20) and operations (page 21). Participant attribution and final sharing confirmation remain team-owned. No video or social-media post is required.

Figures are original vector diagrams or dated screenshots. Crops improve readability without altering records. Links use the frozen evidence commit. This report does not award or promise a score.

Report build uses ReportLab, Pillow and PyMuPDF in an isolated tool environment. Editable source and rebuild instructions are in docs/report. Font: bundled Bitstream Vera. Evidence snapshot: 14 November 2026.